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Asymmetric wettable Janus SiO_2@PDMS/PVA@PVDF membrane for stable membrane distillation

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Highlights

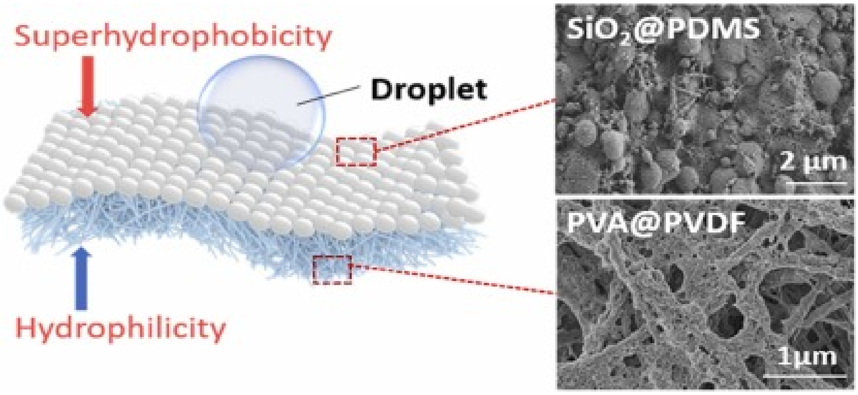
- Janus membrane was prepared by electrospinning and electrospraying.
- The membrane shows asymmetric wettability.
- The superhydrophobic layer consists of SiO_2@PDMS microsphere structure.

- The Janus membrane achieves stable water production in DCMD.

Abstract

Recently, membrane distillation (MD) technology has been widely used in seawater desalination, high-concentration brackish water desalination and industrial wastewater treatment. Conventional electrospinning nanofiber membranes often suffer from low membrane flux and are prone to be contaminated, which weakens the performance of membrane distillation. In this work, we fabricated a Janus membrane (SiO_2@PDMS/PVA@PVDF) with asymmetric wettability (superhydrophobic/hydrophilic) for direct contact membrane distillation (DCMD). The SiO_2@PDMS (SHPo-SP) layer displays excellent superhydrophobicity, which prevents the decrease in salt interception caused by membrane infiltration with brine. The hydrophilic PVA@PVDF (HPi-PP) layer allows for rapid spreading of the liquid over the surface, while accelerating water vapour transport and increasing the water flux in DCMD. Compared to PVDF membrane, the Janus membrane shows long term stable high salt interception (99.995 %) and permeate flux (26.96 kg m^{-2} h^{-1}), which is 66 % higher than that of PVDF membrane. Moreover, the effects of different feed-side brine temperatures, flow rates, and concentrations on DCMD performance have also been explored. This work presents a simple and scalable method for the preparation of asymmetric wettable Janus membranes, which may provide valuable insights for the development of efficient and stable MD technology.

Graphical Abstract



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Keywords

Janus nanofiber membrane; Electrospinning; Electrospraying; Asymmetric wettability; Membrane distillation

1. Introduction

The rapid development of industry and constant increase of the global population have led to a growing global shortage of freshwater resources [1], [2], [3], [4]. In order to increase global freshwater reserves, seawater desalination technologies have been widely developed, including multistage flash evaporation [5], [6], multi-effect distillation [7], reverse osmosis [8], [9], membrane distillation [10] technology, *etc.* Among them, membrane distillation technology is characterized by low cost, low energy consumption and wide range of application, which significantly contribute to the development and utilization of seawater resources [11], [12], [13], [14].

Membrane distillation (MD) is a desalination technology that combines membrane separation process and distillation process [15], [16], [17]. Two liquids with different temperatures separated by a hydrophobic membrane are evaporated and condensed driven by the difference in vapour pressure between two sides of the membrane [18], [19], [20]. The other components of the feed liquid are blocked by the hydrophobic membrane, thus realizing the separation and purification of seawater [10]. Direct contact membrane distillation (DCMD), in which the condensate-side permeate is in direct contact with the membrane, is one of the most fundamental MD technologies. The simplicity of the membrane module and the large trans-membrane flux allow it to be widely used in the desalination of highly saline seawater [21], [22], [23].

The hydrophobicity of the membrane is essential to achieve long-term stable DCMD with the liquid cannot enter the pores to wet the membrane due to surface tension but forms a liquid-vapour interface, which prevents the salt interception rate decrease caused by membrane contamination [24], [25]. In recent years, hydrophobic polymers such as polyvinylidene difluoride (PVDF) [26], polypropylene (PP) [27], Polytetrafluoroethylene (PTFE) [28], *etc.* have become ideal DCMD membrane materials. Commonly used methods for the preparation of membrane materials include non-solvent induced phase transition, melt stretch sintering and electrospinning. Zhu et.al [29] prepared hydrophobic PTFE hollow nanofiber membrane by paste melting, extrusion, stretching and sintering. The membrane surface exhibited hydrophobicity with a contact angle (CA) of ~128° and the effects of feed temperature, vacuum pressure and salt concentration on the MD performance of the hydrophobic PTFE hollow nanofiber membranes were investigated simultaneously. In order to further improve the hydrophobicity, some studies have introduced nanoparticles such as TiO_2 and SiO_2 or hollow nanofibers into the surface of membrane materials to obtain larger roughness. For example, Zhang et al. [30] prepared PVDF flat membranes using a non-solvent induced phase transition method. The hydrophobicity of the PVDF substrate was improved by spraying a mixture of PDMS and hydrophobic SiO_2 nanoparticles on the PVDF surface to achieve a contact angle at ~156°. However, the PVDF flat sheet membrane prepared by this method had a relatively small porosity of 68 %, which is unfavorable for membrane distillation. Xu et.al [31] produced a superhydrophobic membrane by modifying the surface of PP hollow nanofiber membrane with nano-silica and fluorinating. The membrane surface showed a contact angle as high as 157°, and the permeate flux was significantly improved compared with that of ordinary PP membranes. While a significant amount of previous work has been able to achieve contact angles of ~150°, there is limited work on achieving higher contact angles (>160°) [32], which usually requires appropriate methods to build a more uniform roughness while ensuring higher porosity of the polymer membrane. In addition, the gas-liquid interface of a single hydrophobic membrane increases mass transfer resistance and reduces permeate flux during water vapour transport across the membrane to the condensing side [33], [34]. Therefore, the combination of hydrophobic and hydrophilic membranes, i.e. hydrophobic on the feed side and hydrophilic on the condensate side, ensures higher salt interception and permeate flux. In 2013, Prince et. al [35] prepared a three-layer hydrophilic/hydrophobic composite membrane for MD by wet casting and electrospinning for the first time. Compared with hydrophilic membranes, the incorporation of hydrophobic nanofiber layers can effectively improve the salt interception rate and service life. Huang et.al [36] prepared a Janus membrane consisted of Poly (vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP) and nanoparticle polymer composites by electrostatic spinning, chemical vapour deposition and spraying techniques. The Janus membrane features hydrophilic in air and oleophobic under water, which can effectively solve the problem of performance degradation caused by oil contamination. However, asymmetric wettability polymer membranes reported in previous work often utilize different polymers to construct the hydrophobic and hydrophilic layers, which may lead to interfacial exfoliation caused by mismatched thermal expansion coefficients of the polymers [37]. Moreover, the reported methods of polymer membrane preparation often face several challenges [38], [39], [40]. For example, the melt sintering method is complex, less efficient and costly. Additionally, the polymer membranes produced by this method tend to be less thermally stable, making them unsuitable for prolonged membrane distillation [29], [41]. Polymer membranes prepared by the phase conversion typically exhibit low roughness, making it difficult to achieve higher hydrophobicity. This limitation may result in membrane contamination that weakens the permeate flux and salt interception of Janus membranes [39], [42], [43]. In contrast, electrospinning technology, with its low cost, high controllability and scalability, has attracted much attention in recent years. Nanoparticles can be incorporated during or after the electrospinning process to achieve superhydrophobicity on the membrane surface [44]. Besides, comparing to the phase conversion method, electrospinning produces membranes with higher porosity, which facilitates higher water flux [45].

In this work, we present a novelty strategy to fabricate an asymmetric wettable Janus membrane consist of SiO_2@PDMS layer and PVA@PVDF nanofiber membrane for DCMD. The uniform and suitable roughness formed by nano-SiO_2 and PDMS (hydrophobic material) enables one side of the PVDF membrane to exhibit excellent superhydrophobicity with a contact angle of up to 160°, which provides long term resistance to wetting. At the same time, the hydrophilic material PVA was introduced on the other side of the PVDF membrane and crosslinked to obtain a hydrophilic layer, resulting in the conversion of hydrophobicity to hydrophilicity with a contact angle of <5°. Instead of constructing a separate PVA hydrophilic layer, the PVDF nanofibers were wrapped by PVA, which implies that the hydrophilic layer is also based on PVDF and avoids interfacial exfoliation. Moreover, we studied the effect of inlet temperature, feed flow rates and feed concentrations on the performance of DCMD and conducted the long-term stability tests on the Janus membrane.

2. Experimental section

2.1. Materials

Polyvinylidene difluoride (PVDF) was provided by Shanghai Dibo Biotechnology Co., Ltd (China). N, N-dimethylacetamide (DMAC) were obtained from Shanghai Hushi Laboratory Equipment Co., Ltd (China). Polyvinyl alcohol (PVA) and lithium chloride (LiCl) was purchased from Shanghai Aladdin Biochemical Technology Co., Ltd (China). Polydimethylsiloxane (PDMS) was offered by Dow Corning Co., Ltd (USA). Tetrahydrofuran (THF), glutaraldehyde (GA), hydrochloric acid (HCl) and acetone were obtained by Sinopharm Chemical Reagent Co., Ltd (China). Sodium chloride (NaCl) and silicon dioxide (SiO_2, 500 nm) were supplied by Shanghai Macklin Biochemical Technology Co., Ltd (China).

2.2. Preparation of Janus SiO_2@PDMS/PVA@PVDF (Janus SP-PP) membrane with asymmetric wettability

Fabrication of PVDF membrane: The Janus SiO_2@PDMS/PVA@PVDF (Janus SP -PP) membrane with asymmetric wettability was prepared by electrospinning and electrospraying (SS-2535H, Ucalery, Beijing, China) technology (Fig. 1). Firstly, PVDF and LiCl were mixed in to DMAC (PVDF: LiCl: DMAC are 10.9:0.1:89 wt%) and stirred magnetically at 70°C for 6 hrs to obtain an electrospinning solution. Then the prepared solution was loaded into a 10 ml syringe (needle type: 18 G) and electrospun at a positive voltage of 28 kV and a negative voltage of 2.5 kV, with an injection speed of 0.06 mm min^{-1}. The electrospinning distance was 20 cm and the roller contact speed was 50 rpm. The air temperature and humidity were 28 ± 1°C and 50 ± 2 %. The PVDF membrane was obtained after electrospinning for 8 hrs and drying at 70°C for 6 hrs.

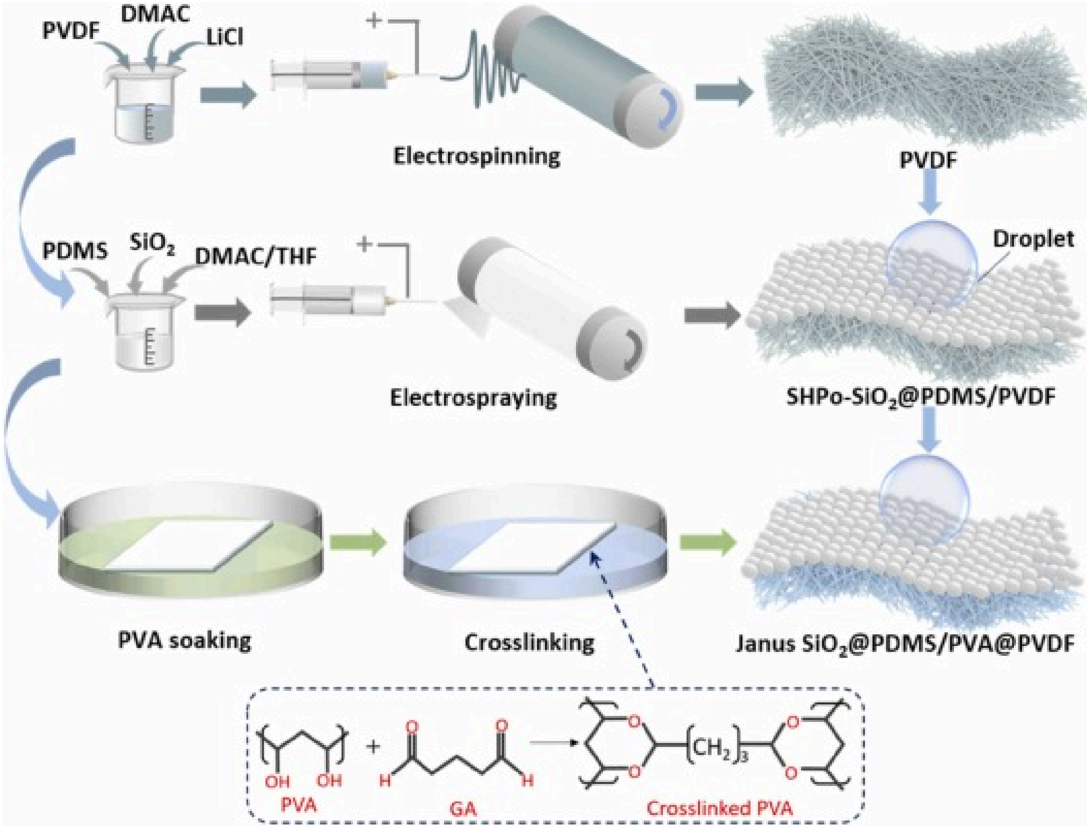

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Fig. 1. Schematic of the preparation process of the Janus SiO_2@PDMS/PVA@PVDF (Janus SP-PP) membrane.

Fabrication of superhydrophobic SiO_2@PDMS (SHPo-SP) layer: Firstly, DMAC and THF were mixed in a mass ratio of 1:1 as solvents (DMAC/THF). A certain amount of SiO_2, PDMS and PVDF were weighted into the DMAC/THF (SiO_2: PDMS: PVDF: DMAC/THF are 0.2:2:3:94.8 wt%) and stirred magnetically at 60℃ for 4 hrs. Then the solution was electrosprayed into the PVDF membrane by a 10 ml syringe (needle type: 24 G). The positive and negative voltage were 14 kV and 2.5 kV, respectively, with the injection speed of 0.15 mm min^{-1}. Moreover, the electrospraying distance was shortened to 9 cm with the roller contact speed unchanged. Finally, after eletrospraying for 1 h, the membrane was dried at 70°C for 4 hrs and hot pressed at 100°C for 2 hrs to enhance the adhesion between the SiO_2@PDMS and the PVDF membrane. Different amounts of SiO_2 nanoparticles were electrosprayed on the PVDF membrane by varying the SiO_2 contents to 0, 0.2, 1, and 2 wt% to investigate the effect of SiO_2 content on the surface superhydrophobicity.

Hydrophilization of PVDF membrane: PVA is a water-soluble polymer, which can impart hydrophilicity to materials through alcoholysis and thus it was used to hydrophilize the hydrophobic PVDF membrane to prepare Janus membrane with asymmetric wettability. Firstly, 1 wt% PVA was dissolved in deionized water and stirred magnetically at 80℃ for 2 hrs to prepare a PVA solution.

Preparation of cross-linking agent: 0.1 wt% GA and 0.5 wt% HCl were added to deionized water, stirred at room temperature and stored for use. Then, the PVDF side of the Janus membrane was immersed in the prepared PVA solution for 1 h at 60°C, and then dried for 1 h at 60°C. Finally, the membrane was immersed in the crosslinking agent for 1 h in a room temperature environment and dried at 80°C for 10 min to enhance the crosslinking. The crosslinking mechanism of GA and PVA can be seen in Fig. 1. Uncross linked PVA is inherently unstable and easily soluble in water. However, the reaction of GA with the large number of hydroxyl groups in PVA can reduce the solubility of PVA while simultaneously imparting hydrophilicity to the hydrophobic PVDF membrane.

2.3. Characterization

The surface morphology of the Janus membrane was detected by field emission scanning electron microscope (FEI 400 FEG, USA and Gemini SEM 500, GER). The element compositions were analyzed by a fourier transformed infrared spectrometer **(**FTIR, Nicolet IS5) and the pore size distribution was measured by a capillary pore size distribution meter (Porolux 1000, IB-FT GmbH Berlin, GER). The water contact angles (CA) were measured by a contact angle goniometer (Ram é-hart 290-U1, USA) with a deionized water droplet of 10 μL. The values of the contact angle are the average of the contact angles measured at 5 different locations on the membrane surface.

The continuous dynamic images of water droplets bouncing on the membrane surface were recorded using a high-speed camera (Phantom v1212, Vision Research, USA).

2.4. Direct contact membrane distillation (DCMD) test

The membrane distillation performance of the Janus SP-PP membrane was tested on a self-designed experimental platform (Fig. 2). (1) DCMD component: The DCMD component was assembled from acrylic sheets (a highly insulating material used to reduce heat loss) and divided into hot side and cold side. Baffles were installed on the hot side to improve the internal flow pattern and increase the thermal efficiency of the system. The DMCD was tested with an effective membrane area of 5 × 10 cm^2, which is consistent with the size of the baffle area. (2) Circulating water system: With different types of working fluid on both sides of the membrane module, the platform can be divided into feed side and permeate side for membrane distillation. Both the feed side and permeate side water circulation were driven by peristaltic pumps. The brine was heated through the heating chamber and then flowed into the hot side of the membrane module for distillation, part of the water vapour entered the cold side of the membrane module through the membrane holes, and the excess brine flowed back to the brine storage tank on the feed side from the outlet of the hot side, and the whole process was carried out in a cycle. Desalinated water production in the cycle was measured by an electronic balance on the permeate side. (3) Data acquisition system: Desalinated water production in the cycle was measured by an electronic balance on the permeate side. Temperature and pressure were recorded by temperature and pressure sensors at the inlet and outlet of the feed-liquid side and permeate side. A conductivity collector was used to monitor and collect solution concentration in real time. Temperature and pressure were collected by temperature and pressure sensors at the inlet and outlet of the feed solution side and permeate side. A conductivity collector was used to monitor the real-time solution concentration. Notably, before conducting the DCMD test using this platform, the entire system needs to be run for ~60 min until it stabilizes. In addition, the tubing should be thoroughly cleaned with DI water before each test to ensure appropriate conductivity on both sides of the membrane. The membrane distillation permeate flux J (kg m^{-2} h^{-1}) was calculated by measuring the mass change of water in the beaker on the permeate side,

$$J = \frac{\Delta m}{A \bullet \Delta t} \tag{1}$$

where Δm is the mass change of water in the beaker, A is the effective area (m^{-2}) of membrane distillation and Δt is the membrane distillation time (h). The salt interception rate R (%) can be calculated by:

$$R = \left(1 - \frac{C_p}{C_f}\right) \times 100\% \tag{2}$$

where C_p and C_f are salt concentrations (g/L) on the permeate and feed sides, respectively. The conductivity of the deionized water on the permeate side was 0.2 μS cm^{-1}. In order to ensure high salt interception rate during the experiment, the conductivity on the permeate side was guaranteed to be ≤ 5 μS cm^{-1}.

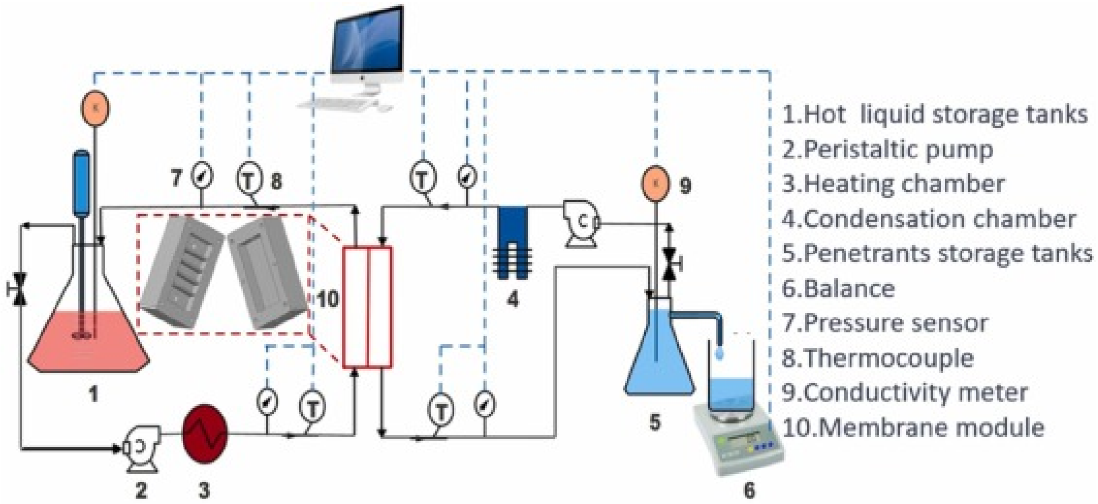

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Fig. 2. Schematic diagram of the test bench for direct contact membrane distillation (DCMD).

3. Results and discussion

3.1. Characterization of the Janus SiO_2@PDMS/PVA@PVDF membrane

The surface and cross-sectional morphology of the Janus SP-PP membrane have been observed by SEM (Fig. 3). As can be seen in Fig. 3a, PVDF membrane consists of a large number of nanofibers with an average diameter of about 400 nm, which are interlaced and stacked to form numerous micron-sized pores. The hydrophobic nature of the PVDF material allows the membrane surface to be slightly hydrophobic, with a contact angle of 132°.The superhydrophobic layer is constructed by electrostatic spraying SiO_2/PDMS/PVDF solution on one side of the PVDF membrane. The spraying solution was dispersed into a large number of small droplets under the force of electrostatic field after spraying from a metal needle, and then the solvent evaporated after the droplets were shot to the receiving end to form rough SiO_2@PDMS microspheres (Fig. 3b). To determine the appropriate SiO_2 concentration when preparing the superhydrophobic SiO_2@PDMS (SHPo-SP) layer, the SiO_2 concentration in the spraying solution was varied to 0.2 wt%, 1.0 wt% and 2.0 wt%. Table 1 show s the contact angle After the deposition of 0.2 wt% SiO_2

via spraying, the contact angle significantly increases to 160°, demonstrating excellent superhydrophobicity. After the SiO_2 concentration was increased to 1.0 wt%, the contact angle was 162°, which was only a small increase, and due to the excessive SiO_2 content, the adhesion decreased and was easy to fall off from the surface. When the SiO_2 content increased to 2 wt%, the surface contact angle decreased to 151°. This decline may due to the agglomeration of excess SiO_2 on the PVDF film surface, which reduces the surface roughness. Therefore, the SiO_2 concentration of 0.2 wt%. The spray solution with 0.2 % SiO_2 concentration was used for the preparation of typical samples. Moreover, the small amount of PVDF added to the spray solution created interconnected nanofibers between the microspheres, which may enhance the adhesion between the microspheres and the PVDF membrane. The hydrophilic PVA@PVDF layer was obtained by immersing the PVDF base membrane in a PVA solution and crosslinking by GA solution. As shown in Fig. 3c, the hydrophilized PVDF nanofibers are wrapped by the hydrophilic polymer PVA, and the surface of nanofibers is rougher and exhibits hydrophilicity. The obtained Janus SiO_2@PDMS/PVA@PVDF (Janus SP-PP) membrane showed hydrophilicity on the PVA@PVDF side and superhydrophobicity on the SiO_2@PDMS side, exhibiting significant asymmetric wettability. The cross-sectional SEM in Fig. 3d shows that the thickness of the Janus membrane is ~80 μm and has a clearly hierarchical structure, with a superhydrophobic SiO_2@PDMS layer at the top and a hydrophilized PVA@PVDF layer at the bottom. Fig. 3e-f show~~s~~ the pore size distribution of the PVDF base membrane and Janus SP-PP membrane. As can be seen, the average pore size of the Janus membrane after the composite of superhydrophobic and hydrophilic layers is 684.7 nm, with a high porosity of 80.2 %, which is still sufficiently suitable for water transport and vapour escape, although it has decreased compared to that of the PVDF membrane (86.6 %).

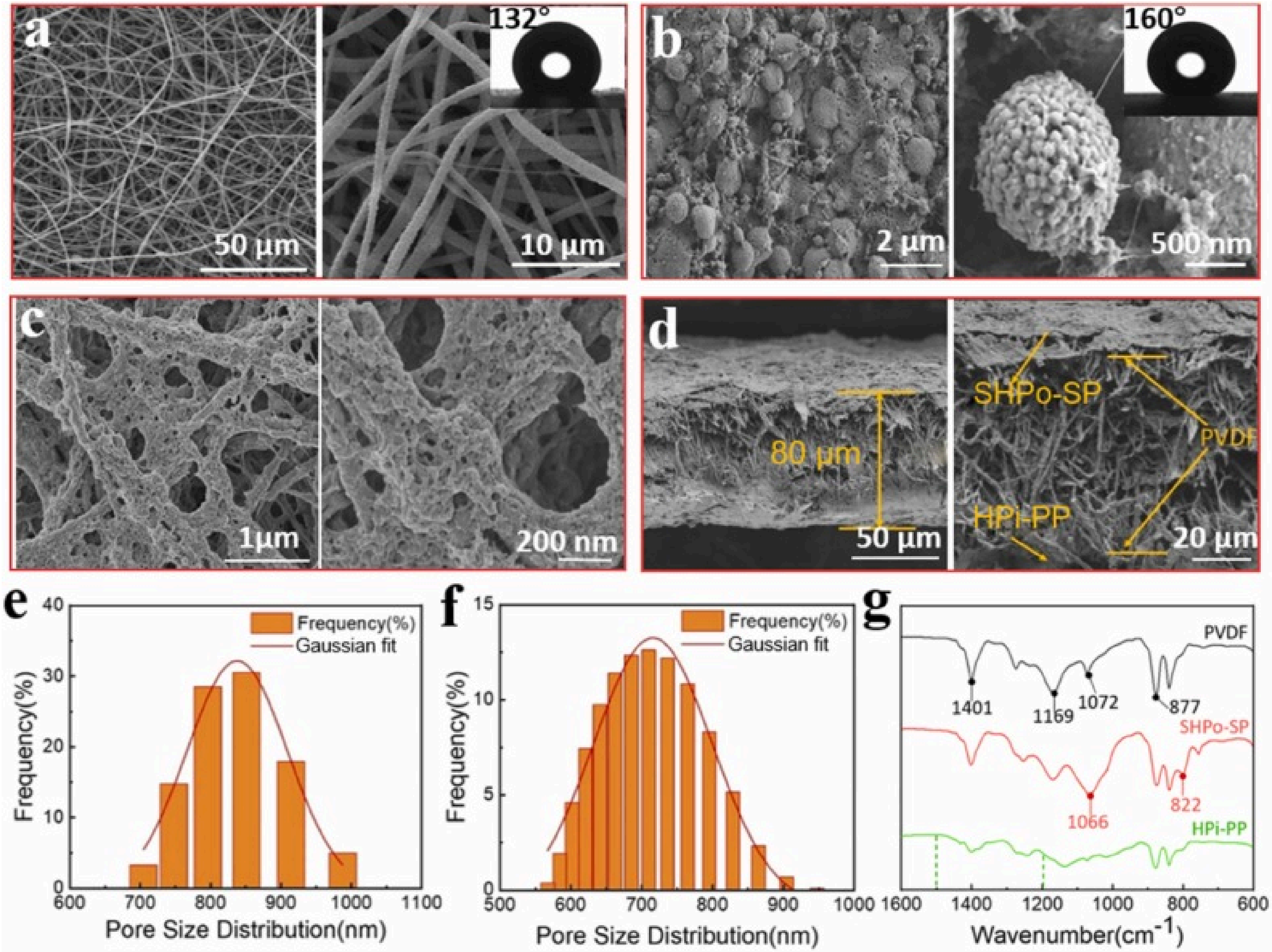

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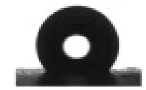
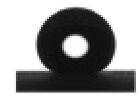
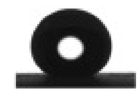
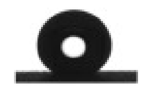
Fig. 3. Surface morphology of the Janus SP-PP membrane. The scanning electron microscopy (SEM) images of the (a) PVDF membrane, (b) superhydrophobic SiO_2@PDMS layer and (c) hydrophilic PVA@PVDF membrane at various magnifications. (d) The

cross-sectional SEM images of the Janus SP-PP membrane. Pore size distribution of the (e) PVDF base membrane and (f) Janus SP-PP membrane. (g) FTIR of the PVDF, SHPo-SP and HPi-PP.

Table 1. Contact angle of the SiO_2@PDMS (SHPo-SP) layer with different SiO_2 content.

SiO_2 concentration/wt%	Contact angle/°	Droplet image
0	132	
0.2	160	
1	162	
2	151	

In order to further investigate the chemical composition of the Janus membrane and the mechanism of superhydrophobicity, FTIR of the Janus membrane was measured and compared with that of the PVDF base membrane. It can be seen that the SHPo-SP, HPo-PP and PVDF membranes all have an absorption peak at 877 cm^{-1}, which indicates the backbone vibration of the C-C bonds. The absorption peak at 1401 cm^{-1} is attributed to the symmetric stretching of -CF_2 bonds in PVDF thus present in SHPo-SP and PVDF membrane[46]. For the SHPo-SP membrane, the absorption peak exhibited at 1169 cm^{-1} reveals the asymmetric stretching and deformation of CH_3 in Si-CH_3 in PDMS and the absorption peak at 820 cm^{-1} represents the Si-$(CH_3)_2$ stretching vibration of PDMS[47]. The pronounce absorption peak at 1022 cm^{-1} is induced by the asymmetric stretching of Si-O-Si, demonstrating that the SHPo-SP layer is composed of PDMS and SiO_2. In the HPi-PP layer, the absorption peak at 1735 cm^{-1} is the characteristic absorption of carbonyl double bond C=O, which is attributed to the residual of the cross-linking agent glutaraldehyde (GA) during the preparation process. Furthermore, the absorption peaks in the range of 1200 cm^{-1}-1500 cm^{-1} originated from the bending vibration of the -OH, which indicated that a PVA hydrophilic layer was successfully coated and crosslinked on the PVDF membrane by acid-catalyzed crosslinking of glutaraldehyde[48].

The asymmetric wettability of the Janus SP-PP membrane was tested using 10 μL droplets impinging on the surface and compared to PVDF membranes. The complete process of droplet spreading, shrinking and bouncing occurs from the initial impact of the droplet on the surface of the membrane at a speed of 1.4 m/s to complete rest, which is recorded by a high-speed camera. As shown in the Fig. 4b, the maximum diameter of the droplet spreading (3 ms) after impacting the SHPo-SP membrane is smaller than that of the PVDF membrane, which is resulted from the superior superhydrophobicity of the SHPo-SP membrane. After that, the droplet contracted and bounced completely, during which the droplet was broken and reached the highest point at 36.6 ms, then fell back again and bounced again, and after bouncing several times, the droplet was completely stationary at 222 ms. However, the PVDF membrane surface only experienced partial bouncing of the broken droplets, and the semi-bouncing droplets reached the highest point at 13.6 ms, then fell back and reached complete rest at 53.4 ms, this phenomenon further illustrates the superhydrophobic property of SHPo-SP layer. For the HPi-PP layer, the droplet contracted after complete spreading at 4.6 ms and was completely stationary at 36.2 ms with a CA of < 5°. However, the contraction transitions of the droplets were actually close to the fully stationary transitions at 10 ms, indicating that the PVA@PVDF layer displays a wettability opposite to that of the SHPo-SP layer.

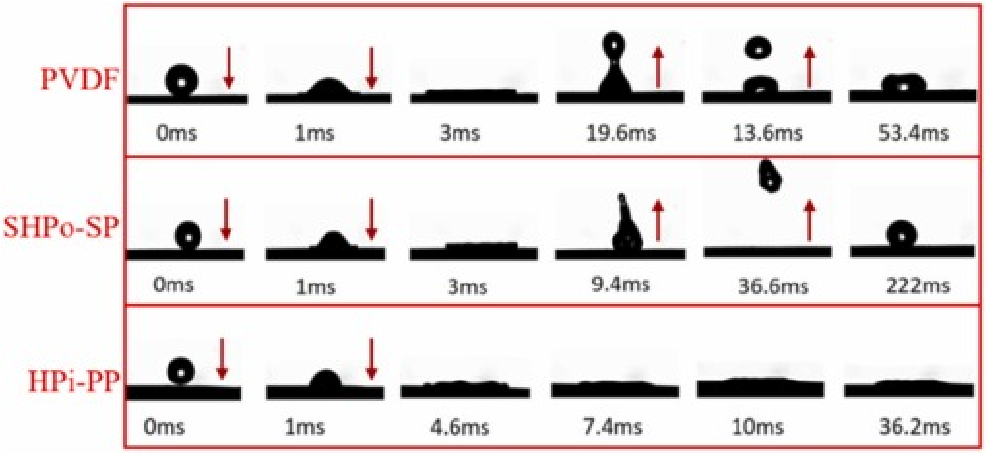


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Fig. 4. Dynamics of droplet impact on PVDF and Janus SP-PP membranes, respectively.

3.2. Factors (inlet temperature, feed flow rates and feed concentrations) affecting DCMD performance of the membrane

The effects of different inlet temperature, feed flow rates and feed concentrations on DCMD performance of the Janus SP-PP membrane have been investigated by a self-designed experimental platform. The effect of feed temperature on the DCMD performance of Janus SP-PP membrane was investigated by varying the feed temperature while maintaining the feed-side brine inlet flow rate of 0.15 m/s and concentration of 3.5 wt%, and the permeate-side deionized water inlet temperature of 20°C and inlet flow rate of 0.15 m/s. Fig. 5a-c exhibit the effect of increasing the inlet temperature from 60°C to 80°C on the DCMD performance of the Janus membrane. As the feed brine temperature increased from 60°C to 75°C, the permeate side maintained a conductivity of less than 5 μS cm^{-1} (Fig. 5a) and a high salt interception rate of 99.995 % (Fig. 5b). Whereas, as the temperature increased to 80°C, the permeate side conductivity kept increasing to 20 μS cm^{-1} and the salt interception decreased to 99.965 %, which could be ascribed to membrane wetting or fouling caused by the excessively high temperature. Fig. 5c shows that the permeate flux of the Janus SP-PP membrane increased and remained stable as the feed temperature increased from 60°C to 70°C (average permeate flux: 60°C, 18.79 kg m^{-2} h^{-1}, 65°C:23.92 kg m^{-2} h^{-1}, and 70°C: 26.48 kg m^{-2} h^{-1}), which is due to the fact that a larger temperature difference between the two sides of the membrane caused by the higher feed temperature, resulting in a larger vapour pressure difference between the two sides of the membrane to facilitate the escape of vapour. However, as the temperature continues to rise, especially to 80°C, the average permeate flux increases up to 47.16 kg m^{-2} h^{-1} and then gradually decreases to a lower level ($<$ 10 kg m^{-2} h^{-1}), indicating that excessive feed temperatures may lead to unstable permeate fluxes.

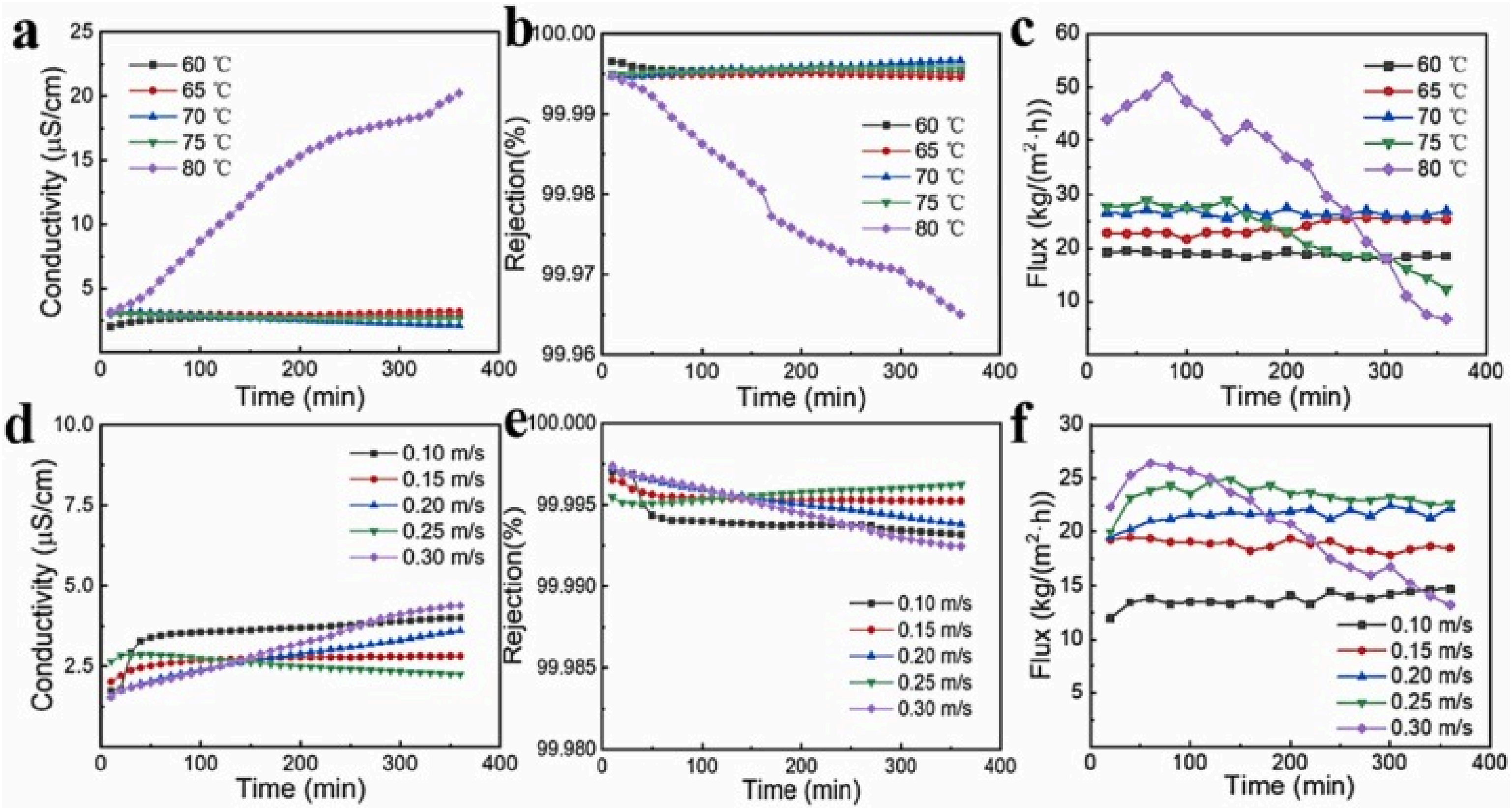

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Fig. 5. The changes of (a) conductivity on the permeate side, (b) salt interception rate and (c) permeate flux of Janus membrane distillation at different inlet temperatures. The changes of (d) conductivity on the permeate side, (e) salt interception rate and (f) permeate flux of Janus membrane distillation at different feed flow rates (n = 1).

To investigate the effect of feed-side inlet flow rate on the performance of DCMD, tests were conducted by increasing the feed-side inlet flow rate from 0.1 m s^{-1} to 0.3 m s^{-1} (in increments of 0.05 m s^{-1}) and keeping the inlet temperature of the feed-side at 60°C, the brine concentration at 3.5 wt%, and the inlet temperature of the permeate-side at 20°C. As shown in Fig. 5d-e, the permeate side conductivity was lower than 5 μS cm^{-1} and maintained a high salt interception > 99.992 % under all working conditions, suggesting that the inlet flow rate had little effect on the salt interception. The permeate flux of the Janus membrane increased with the increasing of the inlet flow rate from 0.1 m s^{-1} to 0.25 m s^{-1} (Fig. 5f), and the flux remained stable at each condition. However, the Janus membrane reached a maximum permeate flux of 26.4 kg m^{-2} h^{-1} at an inlet flow rate of 0.3 m s^{-1}, but then the flux gradually decreased to only 13.19 kg m^{-2} h^{-1}. The reason for the above phenomenon is that at low flow rates, the thicker fluid boundary layer and the stronger temperature polarization effect led to a lower temperature difference and vapour pressure difference on both sides of the membrane, resulting in a lower permeate flux. With the increases of flow rate, the flow conditions within the module improve and the boundary layer thickness and temperature polarization effects decrease while the vapour pressure difference increases, which increases the permeate flux through the membrane, but it is difficult to maintain stability.

The effect of feed-side inlet concentration on the DCMD performance of Janus membranes was explored for salt concentrations of 0 wt%, 3.5 wt%, 5 wt% and 8 wt% (feed-side and permeate-side inlet temperatures of 60°C and 20°C, respectively, and an inlet flow rate of 0.15 m s^{-1}). The conductivity of the permeate side was maintained at $<$ 5 μS cm^{-1} (Fig. 6a) and the salt interception rate was $>$ 99.995 % at all salt concentrations (Fig. 6b), indicating that the salt interception rate of the Janus membrane was insensitive to the change of salt concentration on the feed side, with excellent retention rate for high concentrated brine. In contrast to the effects of feed-side inlet temperature and flow rate on permeate flux, the permeate flux (Fig. 6c) of Janus membrane decreases with the increasing of salt concentration. The average permeates fluxes of Janus membrane at four salt concentration conditions of 0 wt%, 3.5 wt%, 5 wt% and 8 wt% were 20.80 kg m^{-2} h^{-1}, 18.79 kg m^{-2} h^{-1}, 17.15 kg m^{-2} h^{-1} and 15.86 kg m^{-2} h^{-1}, respectively. At a salt concentration of 8 wt% on the feed side, the permeate flux of the membrane remained stable for 2 hrs, after which it gradually decreased to 14.1 kg m^{-2} h^{-1}. Fig. 6d shows the salt crystallization on the surface of Janus membrane after DCMD test at different feed concentration. Comparing to the deionized water (0 wt%) membrane distillation, the Janus membrane surface produced a certain amount of salt crystals after the test at different salt concentrations. It can be attributed to the increase in salt concentration leading to a decrease in vapour partial pressure, which also leads to more sodium chloride crystals being deposited on the membrane surface, in turn leads to a decrease in the effective evaporation area and permeate flux.

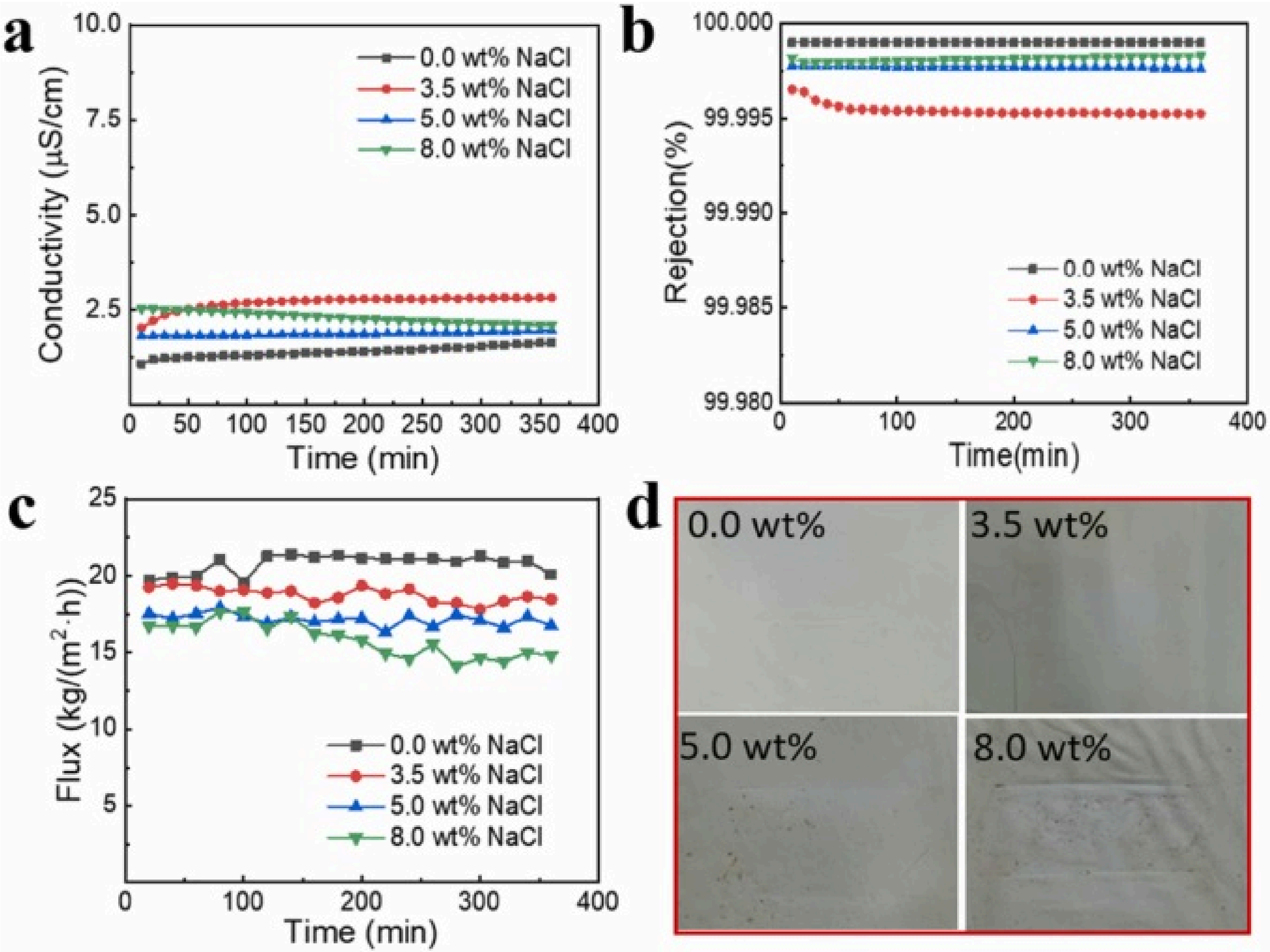

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Fig. 6. The changes of (a) conductivity on the permeate side, (b) salt interception rate and (c) water flux of Janus membrane distillation at different feed concentrations (n = 1). (d) Photographs of the Janus SP-PP membrane surface after the test.

3.3. The stability of Janus SP-PP membrane in DCMD

The stability of Janus SP-PP membrane is a critical criterion for DCMD performance. According to the work in Section 3.2, a stable working condition with high salt interception rate and permeate flux was selected for the long-term stability test of DCMD of Janus membrane and compare with that of the PVDF membrane. The experiments were conducted by choosing a feed-side inlet temperature of 70°C, a brine concentration of 3.5 wt%, and an inlet flow rate of 0.20 m s^{-1} on both sides, and the permeate side conductivity, salt interception rate and permeate fluxes of the Janus SP-PP membrane and PVDF membranes were tested for the 24 hrs DCMD. Fig. 7a-c exhibits the permeate side conductivity, salt interception and permeate flux of Janus membrane and PVDF membrane during 24 hrs. During the DCMD test, the conductivity of permeate side of Janus membrane were always stable at ~1 μS cm-1, and the salt interception rate were more than 99.995 %, which was close to that of deionized water. For the PVDF membrane, the permeate side conductivity increased and the salt interception rate decreased constantly, with the salt interception rate decreasing to 99.960 % at the end of the test. The results revealed that when the brine on the feed side contacts the superhydrophobic side of the Janus membrane, the superhydrophobic SiO_2@PDMS microsphere structure on the surface can increase the air gap between the membrane surface and the liquid, therefore reduce the possibility for wetting of the pores. While the insufficient superhydrophobicity of the PVDF membrane leads to wetting and contamination during prolonged DCMD testing, resulting in a significant drop in retention rate. Fig. 7c compares the permeate fluxes of Janus membrane and PVDF membrane during the long term DCMD test. The PVDF membrane permeate flux gradually decreased with an average flux of 16.28 kg m-2 h-1, whereas the Janus membrane showed a remarkably stable and high flux of 26.96kgm^{-2} h^{-1} over a long period of time, which is 66 % higher than that of the PVDF membrane. In addition, the HPi-PP layer of the Janus membrane on the permeation side enables the deionized water to spread quickly on the surface, with no air gap between the liquid and the surface, shortening the vapour transport distance and accelerating condensation, which is also contributing to the increase in permeation flux. Notably, the introduction of the SiO_2@PDMS layer does not contribute to the increase in flux, as demonstrated in many previous studies [49]. The amount of salt crystallization caused by concentration on the permeate side surface of the Janus membrane was significantly lower than that of the PVDF membrane after 24 hrs DCMD (Fig. 7d), further indicating that the Janus membrane has higher permeate flux and salt interception than PVDF in desalination of 3.5 % concentration of NaCl solution. Moreover, Table 2 compares the wettability, pore size, and DCMD properties of membranes based on PVDF, PDMS, and SiO_2 in selected previous literature. Flat sheet PVDF membranes[30], [50] prepared using the phase inversion method tend to show a narrower average pore size, which may result in a lower flux. In addition, the asymmetric wettability of the membrane reduces the transmembrane mass transfer resistance, resulting in a highly permeable membrane. Even though polymeric membranes with a wetting gradient are often multilayer composites, which tend to have a greater thickness than a single superhydrophobic membrane, they can still exhibit comparable fluxes to superhydrophobic membranes[49].

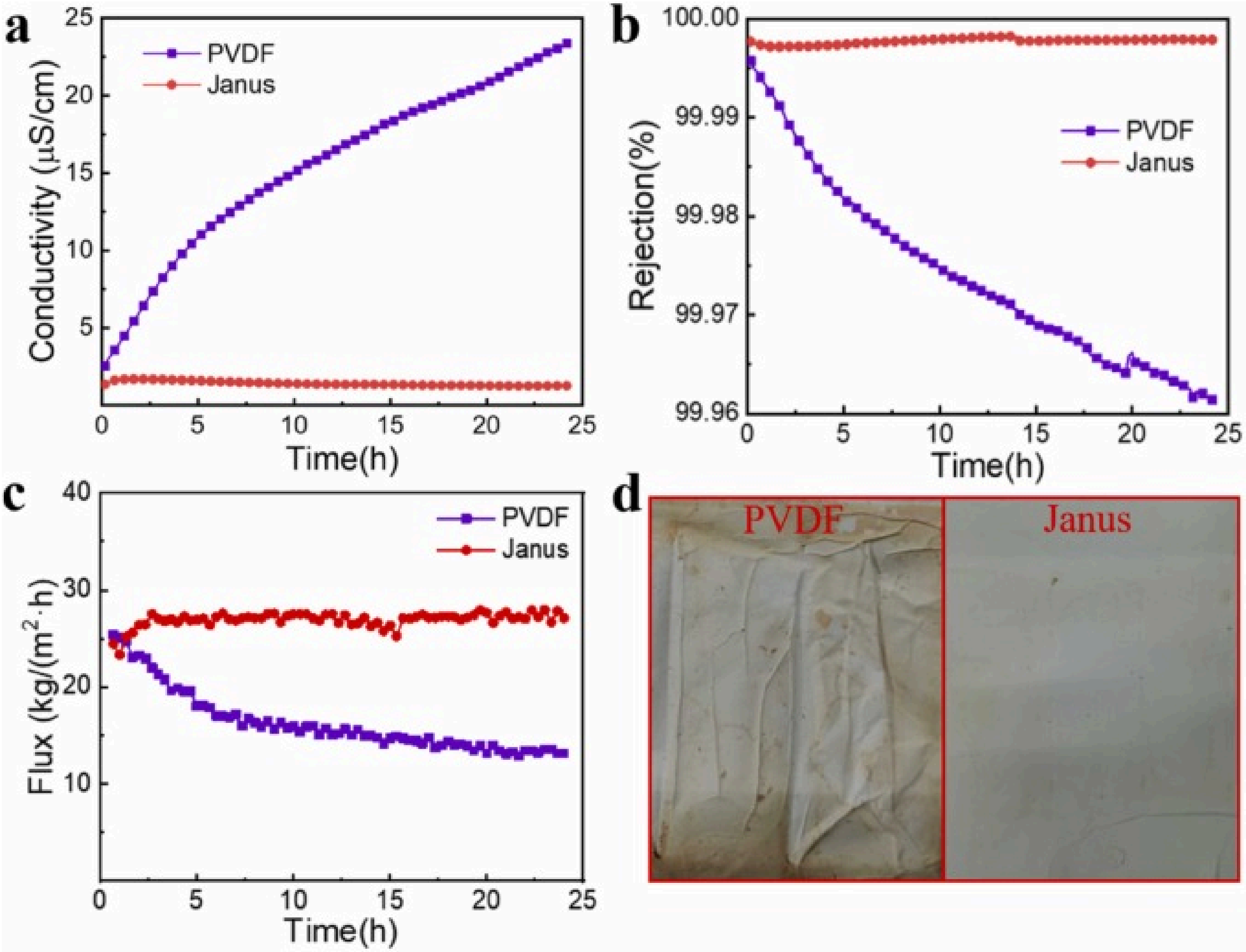

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Fig. 7. The comparison of (a) conductivity on the permeate side, (b) permeate flux and (c) salt interception rate between Janus SP-PP membrane and PVDF membrane during 25 h of membrane distillation (n = 1). (d) Photographs of Janus SP-PP membrane and PVDF membrane after 25 h of membrane distillation.

Table 2. Comparison of the properties of membranes based on PVDF, SiO_2, PDMS, etc. in previous literature with this work.

Wettability	Reference	Mean pore size (μm)	Contact angle (°)	MD parameters		NaCl (wt%)	Flux kg m^{-2} h^{-1}
				$T_{f,in}$ (℃)	$T_{p,in}$ (℃)		
Mono-wettability	[50]	0.103	153	70	15	3.5	27.4
	[51]	1.24	156.4	60	20	~3.0	30.7
	[30]	0.36	156	70	20	3.5	~8
Asymmetric wettability	[26]	1/11 (PH/PAN)	150/100	60	20	3.5	30
	[49]	1.45	164/0	60	20	3.5	28.2
	[52]	0.45	130/87	50	20	2.9	9
	This work	0.685	160/<5	70	20	3.5	27

4. Conclusion

In summary, a Janus membrane with asymmetric wettability have successfully prepared for DCMD by electrospinning, electrospraying and chemical combination methods. Factors affecting water flux and salt interception of Janus membrane in DCMD have been investigated and the long-term stability of Janus SP-PP compared to the common PVDF membrane has been verified. The superior superhydrophobicity of the SiO_2@PDMS (SHPo-SP) layer and the hydrophilicity of the PVA@PVDF (HPi-PP) layer of the Janus membrane with high porosity enable it to maintain a stable high- salt interception rate (> 99.995 %) and water flux (26.96 kg m-2h^{-1}) during the DCMD process, which is 66 % higher than that of the PVDF membrane. In addition, the superhydrophobicity of the Janus membrane effectively reduces the generation of crystalline salts on the surface, resulting in longer-term stability. The concepts presented in this work, especially the exploiting of asymmetric wettability to enhance the DCMD performance and stability of membranes, can provide promising insights for the development of low-cost, scalable and durable seawater desalination technologies.

CRediT authorship contribution statement

Xuemei Chen: Writing – review & editing, Supervision. **Xin Meng:** Writing – original draft, Methodology, Formal analysis. **Wei Wang:** Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Recommended articles

Data availability

Data will be made available on request.

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